MILITARY SPECIFICATION

TABLE, PHOTOINTERPRETER AND ANALYST, ILLUMINATED, SPLIT STAGE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an illuminator for viewing and annotating photographic film negatives and positives for photointerperting and analysis purposes.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NNN-P-40	- Paper, Lens.
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.
MIL-B-26195	- Box, Wood-Cleated, Skidded, Load-Bearing Base.

FSC 6675

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STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, and Blocking.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS26565	- Spool-Photographic Film, Aerial.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The table shall have a two-section viewing surface which shall accommodate one strand of 9.5 inch wide aerial roll film or two strands of film with a combined width of up to 10.5 inches. It shall be equipped with a film transport system that shall facilitate manual transport of the film in both directions. It shall be equipped with an optics carriage which shall provide for mounting and movement of a Bausch & Lomb Stereozoom 240 optical device or equal over the viewing surface. It shall be equipped with external masks which will block extraneous light at front and back of illuminated surface. The table shall be mounted on a stand which can be adjusted to provide a comfortable working height for the operator (see figure 1).

3.2 First article. When specified, (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Weight. The table shall weigh not more than 360 pounds.

3.5 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

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3.6 Rigidity and alignment. Each part of the table shall be of sufficient strength and rigidity to maintain smooth operation and proper alignment.

3.7 Shock, inclination and vibration. The table shall operate under normal conditions and shall withstand shock, vibration and inclination incident to handling.

3.8 Screw threads. Screw threads employed in fabrication of the table shall conform to FED-STD-H28 and shall be of commercial standard size. Coarse threads shall be used in soft metals. Screws self-locking, pan head, full thread when used, shall have locking means.

3.9 Dimensions. The table, with reel brackets installed, but not film spools, shall be not more than 62 inches wide, shall be approximatley 31 inches deep, and not more than 78 inches high.

3.10 Film damage. The table shall not scratch or otherwise damage the film base or emulsion in any manner during its operation.

3.11 Principal parts. The table shall consist of the following principal parts:

(a) Light box
(b) Viewing stages
(c) Light source
(d) Film transport system
(e) Elevating stand
(f) Optics carriage

3.12 Light box. The light box shall enclose the light source to protect against dust, moisture and damage. It shall provide support for the viewing stages and film transport system. It shall be constructed of aluminum alloy and shall not be connected to any part of the electrical circuit. It shall provide a mechanism for separation of viewing stages to accommodate film loop accumulation.

3.13 Viewing stages. There shall be two viewing stages, approximately 11 x 18 inches each. The viewing surface shall have 0.375-inch plate glass for viewing and annotating film transparencies. The plate glass covering shall consist of a clear top sheet as a work surface, and a translucent sub-sheet for diffusion of the light. The viewing surface shall be capable of supporting a weight of 20 pounds with a base approximatley 7 inches in diameter. The viewing surface shall not sag, warp, or crack from expansion or heat generated within the light table, under the operational or environmental conditions specified herein. Means employed to prevent damage shall not cast shadows or otherwise affect the evenness of illumination at the viewing surface.

3.13.1 Viewing area. The viewing area shall be approximately 11 x 38 inches, without any obstruction except for the slot between the stages for the film loop accumulation mechanism.

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3.13.2 Temperature rise. The temperature rise at the viewing surface shall not damage nitrate or acetate base film when viewed continuously over a 4 hour period. When tested as specified in 4.5.2.4, the temperature of the viewing surface shall rise not more than 30° F above ambient temperature.

3.13.3 Illumination. The light at the viewing surface shall be adjustable to any intensity between maximum and minimum. When tested as specified in 4.5.2.2, maximum intensity shall provide not less than 2,200 footlamberts at the center of each viewing stage when the light table is located in an area with an ambient temperature of 65° F to 70° F. When determined as specified in 4.5.2.2, the intensity at any test point in the viewing area shall not vary more than 25 percent from the average intensity. The intensity shall be adjustable down to approximately 500 footlamberts. The color of the light produced shall make a visual match to a source of 4,500 to 5,500 degrees Kelvin when set at maximum illumination.

3.13.4 Masks. Each viewing stage shall be equipped with manually operated masking devices capable of blocking extraneous light at the front and rear of the stage.

3.14 Light source. The light source or sources shall be contained within the housing. Each lamp assembly shall be of a rugged type, mounted in a manner to withstand normal handling and transportation of the table.

3.15 Film transport system. The table shall be equiped with a manually operable film transport system. The system shall facilitate transport of one or two strands of film simultaneously in both directions across the viewing stages.

3.15.1 Reel brackets. The brackets shall accommodate film spools conforming to MS26565 in widths up to 9.5 inches and lenghts up to 1,000 feet. The design and mounting of the brackets shall permit rewinding the 1,000 foot spool with the table on a flat surface. The brackets shall be rigidly attached to both ends of the table, but shall be readily removable. The transport system shall allow for split-vertical viewing of films.

3.15.2 Film rollers. Film rollers shall be mounted on each end of the light box to support the film slightly above the viewing stage surface. The rollers shall be constructed in segments so as to allow transport of a second strand of film at a different speed or in the opposite direction as the first.

3.15.3 Film accumulation mechanism. The table shall be equipped with a manually operable film accumulation mechanism which shall allow for viewing of widely separated targets (up to 75 inches apart) on one strand of film.

3.16 Elevating stand. The main support for the table shall be an elevating stand capable of positioning the viewing surface to any position between approximately 30 and 45 inches above the floor. The stand shall be mounted on four rubber-tired casters, the front two of which shall include locking mechanism to prevent movement of the light table when in operation.

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3.17 Optics carriage. The table shall be equipped with a carriage which shall allow movement of an optical device over the viewing surface in both the X- and Y-direction. Provisions shall be made for movement of the optical device in the Z-direction for purposes of optical instrument focusing. The optics carriage shall allow for collimation of the mount to the stage glass to within plus or minus 5 minutes of arc in the X-direction and plus or minus 5 minutes of arc in the Y-direction over the full travel of the viewing stage and maintain parallelism to the stage glass to within plus or minus 0.005 inch. The carriage shall be equipped with electromechanical clutches which shall lock the carriage position when power is applied to the table. Rapid positioning of the carriage shall be facilitated by a momentary release switch located on the optics mount. Fine-feed knobs located on the carriage rails shall be used for final positioning. The carriage clutch circuit shall be equipped with an on/off switch to allow the operator to operate the carriage without the clutches.

3.18 Electrical requirements. The table shall operate from a 120 volt, 60 Hertz, single phase ac electrical power supply. It shall draw not more than 8 amperes.

3.18.1 Switches. The table shall be equipped with a main power on/off switch that shall control all table circuits. The light sources shall be controlled by a single on/off switch and a single potentiometer for illumination level control. All switches and controls shall be located such that they are easily accessible to the operator.

3.18.2 Wiring. All electrical wiring of the light table shall conform to good commercial practices.

3.18.3 Power cord. The table shall be provided with a 10 foot length of rubber-covered, 3 wire, grounding type power cord. One end shall be permanently attached to the light table and the other end shall terminate in a 3 prong male rubber plug.

3.19 Finish.

3.19.1 Interior. The interior of the lamp housings shall be given a durable white finish.

3.19.2 Exterior. All exterior parts shall be given a durable finish. Aluminum parts shall be treated in accordance with MIL-A-8625. The frame and other untreated exposed metal surfaces shall be given a durable light green finish.

3.20 Identification marking. Identification marking of the equipment shall be in accordance with MIL-STD-130.

3.21 Workmanship. All parts, components, and assemblies shall be clean of harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsbility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 CLassification of inspection. The inspection requirements specified herein are classified as follows:

(a) First-article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first-article tables shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-article tables shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each table shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each table shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The tables shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Weight not as specified.
103. Components missing or not as specified.
104. Treatment and painting not as specified.
105. Material not as specified.
106. Finish and workmanship not as specified.

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4.5.2 Tests.

4.5.2.1 Operation. Operate the controls, reel brackets and switches. Inability of the controls, reels and locks to operate smoothly and without binding, or failure of the lighting system to operate, shall constitute failure of this test.

4.5.2.2 Illumination. Using a Tektronix J16 illumination meter, or equal, measure the light emitted from various parts of the viewing area with the controls set for maximum high intensity. Measure the intensity one inch in from the sides of the viewing area at the four corners, at the mid-points of the four sides, the center and at the four mid-points between center and corners along the diagonals. Visually examine the viewing area for local bright spots or shadows. If any are observed, measure the brightness of that area. Nonconformance to 3.13.3 shall constitute failure of this test.

4.5.2.3 Color of light. Compare the color of the viewing surface of the table at high intensity with a diffuse light source having a color temperature of 4500 to 5500 degrees Kelvin. Nonconformance to 3.14 shall constitute failure of this test.

4.5.2.4 Temperature rise. Attach a wire thermocouple to the center of the viewing surface with transparent plastic tape and on it lay either a 9 by 18 inch picture negative bearing the usual distribution of densities, or a negative uniformly fogged and developed to a density of 0.9 plus or minus 0.1. Operate the table starting at an ambient temperature of 75° F plus or minus 10° F, for four hours. Nonconformance to 3.13.2 shall constitute failure of this test.

4.5.2.5 Viewing surface rigidity. Establish a plane parallel to the viewing surface of the table, and with gauges measure the distance from the datum plane to the glass surface at existing room temperature. Place a circular object seven inches in diameter plus or minus 1/4 inch with a central opening, and weighing not less than twenty pounds, in the center of the viewing surface. Operate the table at high intensity in an ambient temperature of 105° F for 20 hours. Without changing conditions, measure the distance from the datum plane to the center of the viewing surface. A distance variation of more than 0.1 inch shall constitute failure of this test.

4.5.2.6 Viewing surface warping. Establish a plane parallel to the viewing surface of the table, and with gauges measure the distance from the datum plane to the glass surface at room temperature. Raise the ambient temperature to 160° F and maintain for four hours. Allow the table to cool to room temperature. Remeasure the distance from the datum plane to the glass surface. Visually examine the glass surface. A change of distance to any point on the viewing surface of more than 0.05 inch, or any indication of damage, shall constitute failure of this test.

4.5.2.7 Operational. Determine that the various size spools fit easily and securely in the reel brackets. Load a spool with 1000 feet of 9-1/2 inch wide film on the table and determine that the film transport system operates satisfactorily. Examine the film after several rewindings. Evidence of scratching

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or other damage to the film, or evidence that the spools and reel brackets do not operate smoothly and securely shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials and containers not as specified for level A or B. Each incorrect material or containers shall be considered one defect.
108.	Blocking and bracing not as specified for level A or B.
109.	Glass viewing surface not covered with lens paper and fiberboard as specified for level A or B.
110.	Barrier not properly sealed for level A or B.
111.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The preproduction pack may be prepared utilizing a production table.

5.2 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Viewing surface. The viewing surface shall be covered with lens paper conforming to NNN-P-40. The viewing surface shall be cushioned with fiberboard conforming to PPP-F-320, type CF, SW, 175 pounds. The cushioning shall be held in place with tape conforming to MIL-T-22085, type 2.

5.2.1.2 Table assembly. The entire table assembly shall be preserved in accordance with method IIa of MIL-P-116. Moveable components shall be secured as required. Corners and projections shall be cushioned as required to prevent damage to the barrier material.

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5.2.2 Level B. Preservation shall be as specified for level A, except that preservation of table assembly shall be in accordance with method Ia-16 of MIL-P-116.

5.2.3 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.3.1 Level A. The table assembly shall be packed in a box conforming to MIL-B-26195, type II, with rubbing strips and plywood panels. The table assembly shall be blocked, cushioned, and anchored in accordance with MIL-STD-1186.

5.3.2 Level B. The table assembly shall be packed as specified for level A, except that a type I container shall be used.

5.3.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.4 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The table is intended for viewing negative and positive film transparencies by transillumination and to provide a flat, smooth surface for marking annotations, overlays and tracings. The table is intended to be installed in Topographic Support System (TSS) vans for photointerpreting and analytical purposes.

6.2 Order data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of the first article and number of units required (see 3.2).
(c) When preproduction pack is not required (see 5.1).
(d) Time frame required for submission of preproduction pack, when applicable (see 5.1).
(e) Degree of preservation and degree of packing (see 5.2 and 5.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first-produced table. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

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6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meet the requirements of this specification (see 3.3).

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the tables in accordance with this specification (see 5.1).

6.6 Acceptable model. It is considered that the Richards Corporation, McLean, Virginia model MIM3-35100, or equal, is acceptable.

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Preparing activity:
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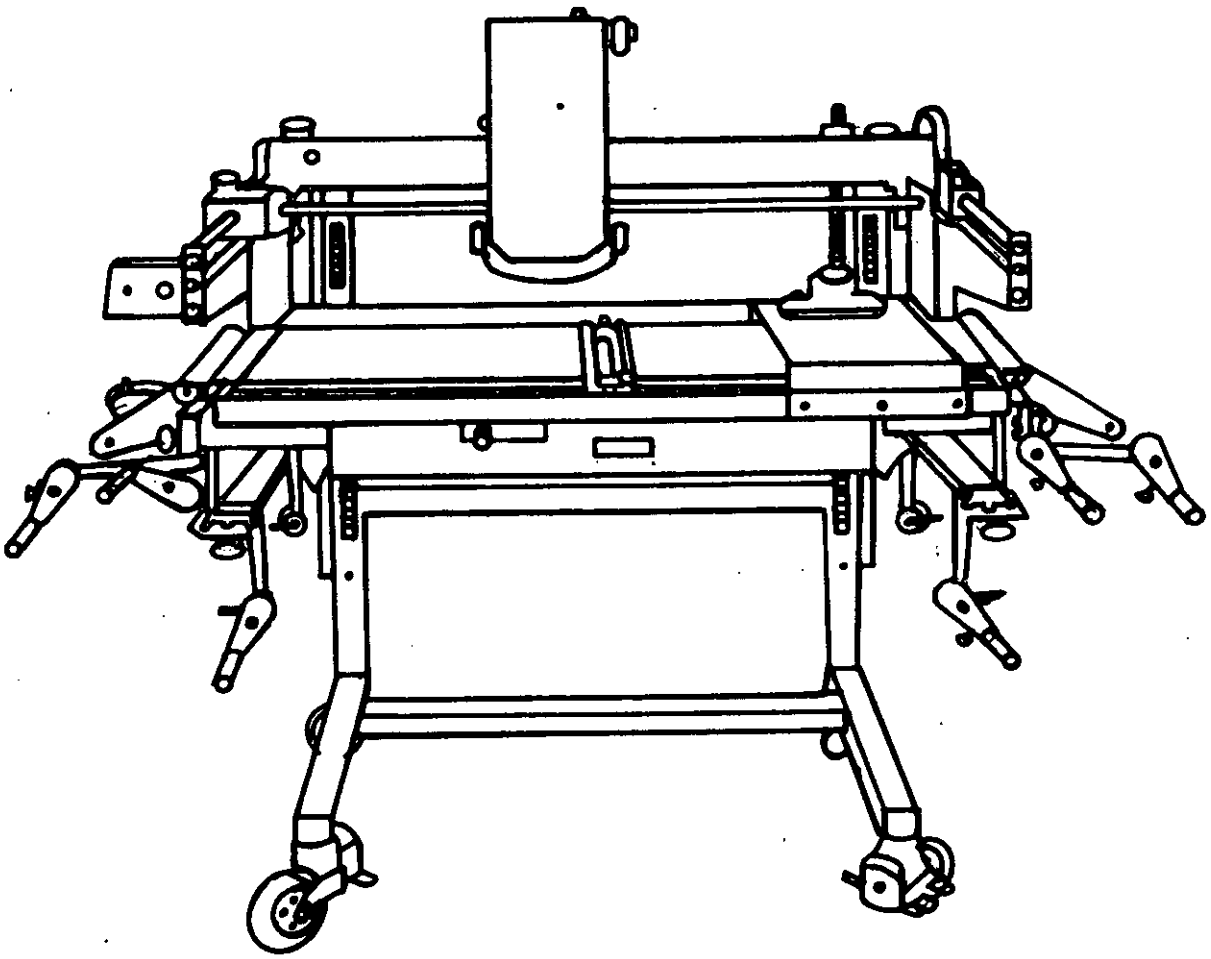


FIGURE 1. Table, photointerpreter.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-52993(ME) Table, Photointerpreter and Analyst, Illuminated, Split Stage

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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